

# Project Report: TaskFlow To-Do Web Application

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## 1. Introduction

TaskFlow is a premium, responsive web application built with Flask and Python. It allows users to manage their daily tasks with full CRUD functionality, interactive modals, and real-time filtering.

## 2. Technology Stack

**Backend:** Flask, SQLAlchemy (SQLite), Flask-Migrate

**Frontend:** Bootstrap 5, Bootstrap Icons, HTML/CSS, JavaScript

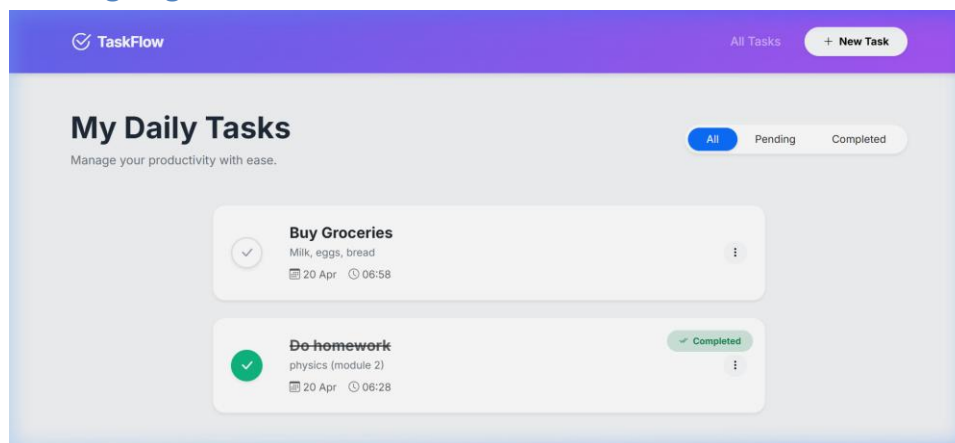
**Deployment:** Vercel, Render, GitHub

## 3. Key Features

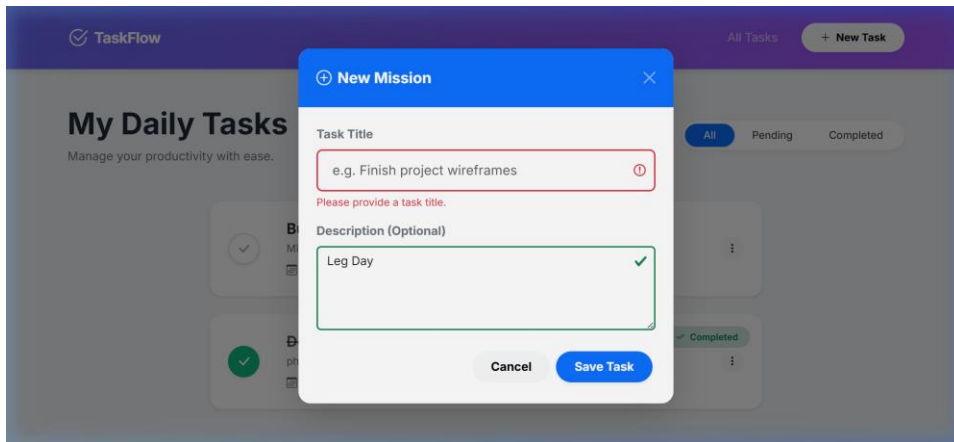
- Create, Read, Update, and Delete tasks.
- Modal-based UI for creating/editing tasks.
- Real-time filtering (All, Pending, Completed).
- Mobile-responsive design with Floating Action Button.
- Database versioning and self-healing migrations.

## 4. Application Interface

### Landing Page

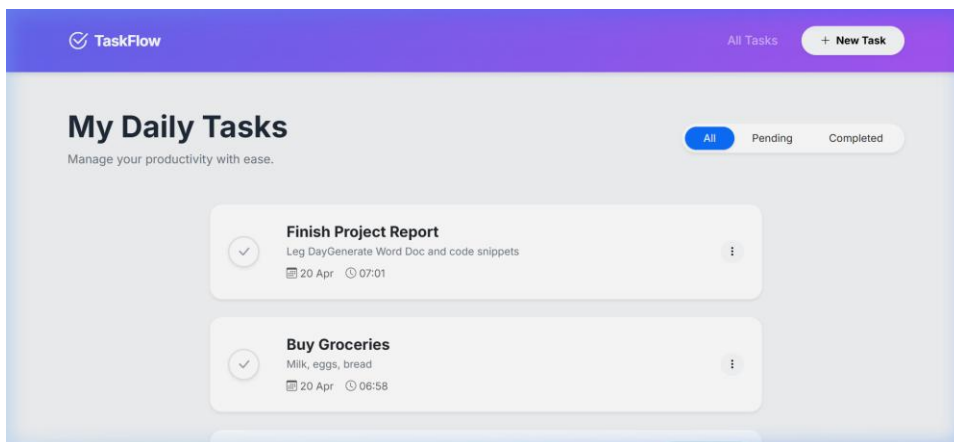


## New Task Modal



The screenshot shows the 'New Mission' modal in the TaskFlow application. The modal is centered over the 'My Daily Tasks' page. It features a blue header with a close button. The 'Task Title' field is highlighted with a red border and a red error message: 'Please provide a task title.' The 'Description (Optional)' field has a green border and a green checkmark. At the bottom, there are 'Cancel' and 'Save Task' buttons.

## Updated Task List



The screenshot shows the 'My Daily Tasks' page in the TaskFlow application. The page has a purple header with the TaskFlow logo and a '+ New Task' button. Below the header, there are tabs for 'All', 'Pending', and 'Completed'. The main content area displays a list of tasks. The first task is 'Finish Project Report' with a description 'Leg DayGenerate Word Doc and code snippets' and a timestamp '20 Apr 07:01'. The second task is 'Buy Groceries' with a description 'Milk, eggs, bread' and a timestamp '20 Apr 06:58'.

## 5. Important Code Snippets

### Flask Backend (app.py)

```
import os
from datetime import datetime
from flask import Flask, render_template, request, redirect, url_for, flash
from flask_sqlalchemy import SQLAlchemy
from dotenv import load_dotenv

# Load .env if present
load_dotenv()

app = Flask(__name__)

# --- CONFIGURATION ---
app.config['SECRET_KEY'] = os.environ.get('SECRET_KEY') or 'dev-secret-key-12345'
```

```

app.config['SQLALCHEMY_TRACK_MODIFICATIONS'] = False

# Handle Vercel's read-only filesystem
if os.environ.get('VERCEL'):
    app.config['SQLALCHEMY_DATABASE_URI'] = os.environ.get('DATABASE_URL') or
'sqlite:///tmp/app.db'
else:
    basedir = os.path.abspath(os.path.dirname(__file__))
    app.config['SQLALCHEMY_DATABASE_URI'] = os.environ.get('DATABASE_URL') or \
'sqlite:/// ' + os.path.join(basedir, 'app.db')

db = SQLAlchemy(app)

# --- MODELS ---
class Task(db.Model):
    id = db.Column(db.Integer, primary_key=True)
    title = db.Column(db.String(100), nullable=False)
    description = db.Column(db.String(255), nullable=True)
    completed = db.Column(db.Boolean, default=False)
    created_at = db.Column(db.DateTime, default=datetime.utcnow)

    def __init__(self, title, description=None, completed=False):
        self.title = title
        self.description = description
        self.completed = completed

# Initialize tables
with app.app_context():
    try:
        db.create_all()
    except Exception as e:
        print(f"Initial DB creation failed: {e}")

# --- ROUTES ---
@app.route('/')
def index():
    filter_...

```

### Configuration (config.py)

```

import os
from dotenv import load_dotenv

basedir = os.path.abspath(os.path.dirname(__file__))

```

```
load_dotenv(os.path.join(basedir, '.env'))

class Config:
    SECRET_KEY = os.environ.get('SECRET_KEY') or 'dev-secret-key-12345'

    # Handle Vercel's read-only filesystem by using /tmp for SQLite
    if os.environ.get('VERCEL'):
        SQLALCHEMY_DATABASE_URI = os.environ.get('DATABASE_URL') or \
            'sqlite:///tmp/app.db'
    else:
        SQLALCHEMY_DATABASE_URI = os.environ.get('DATABASE_URL') or \
            'sqlite:/// ' + os.path.join(basedir, 'app.db')

    SQLALCHEMY_TRACK_MODIFICATIONS = False
```

## 6. Project Links

**GitHub Repository:** <https://github.com/rutujakharode6-max/To-Do-Web-app.git>

**Hosted Link (Vercel Dashboard):** <https://vercel.com/rutujakharode6-maxs-projects>

**Localhost Server:** <http://127.0.0.1:5000>